

Characterization of intron retention events upon loss of the RNA binding protein Nab2 in *Drosophila melanogaster*

Rebecca McSweeney Corbett and Moberg Labs,, BIOS 584 Final Project, Fall 2025

Emory University

INTRODUCTION

Loss of the highly conserved RNA binding protein ZC3H14 has been linked to a rare form of intellectual disability in human patients¹. Loss of the ZC3H14 ortholog, Nab2 in the fruit fly *Drosophila melanogaster* results in reduced viability, lifespan, and locomotion in addition to other neurological phenotypes, providing an excellent model for interrogating the effects of Nab2 loss¹. In addition to altered steady-state levels for hundreds of transcripts, previous RNA sequencing results from the heads of Nab2 mutant flies are consistent with a subset of the mature mRNA transcripts having retained intronic sequences in the Nab2 mutant flies. A systematic characterization of these intron retention events, however, has yet to be performed.

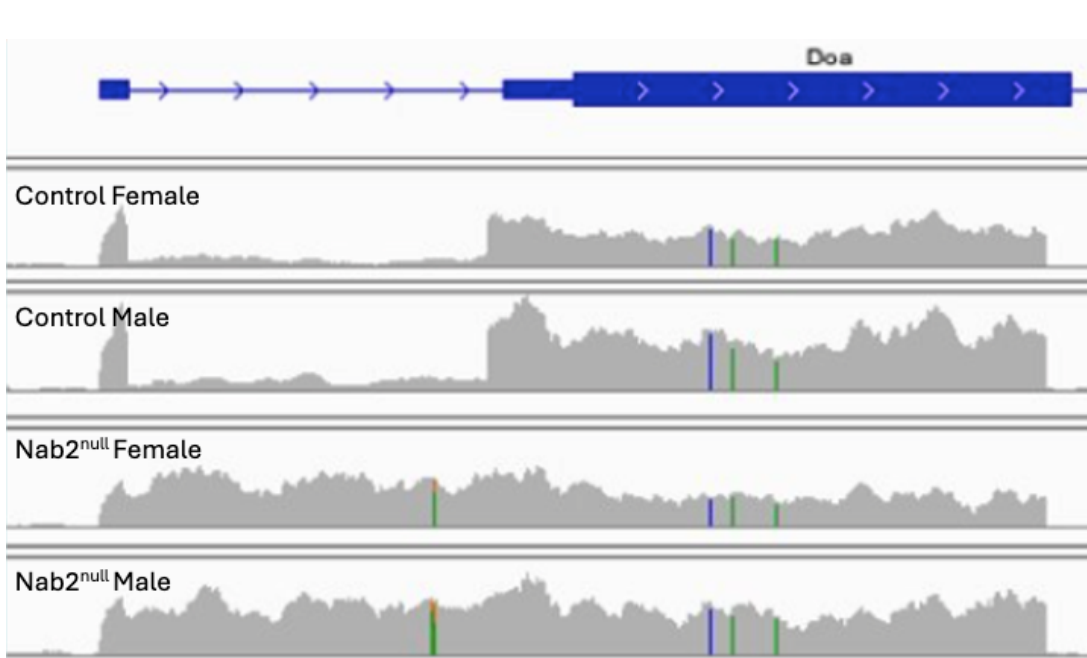


Figure 1: Sequencing results are consistent with intron retention in some mature RNA transcripts in Nab2 mutant flies: This image displays RNA sequencing reads mapped to an isoform of Darkener of apricot (Doa) in wild-type and Nab2 mutant males and females. While the intronic sequence is properly spliced out in the control samples, the Nab2 mutant flies have reads mapping across this intron. Sequencing reads were visualized using the Integrative Genomics Viewer (IGV).

The goal of this work is to characterize the transcript levels and lengths of different transcript features in a Nab2 mutant sample, and to compare transcripts classified by whether or not they contain intronic sequences. We worked to answer the following questions in our analysis:

1. Are there differences in transcript levels for transcripts with and without a retained intron?
2. Do the average transcript lengths differ between transcripts where we observe intron retention events and those we do not?
3. How do the proportion of the overall length of the 5' untranslated region (UTR), CDS, and 3' UTR sequences compare for genes where we observe intron retention events and those we do not?

METHODS

Determining Intron Retention Status and Initial Analyses: Short read RNA sequencing reads were collected from Nab2 mutant female flies and mapped as previously described². We then used featureCounts to calculate the read counts and feature lengths for each gene by inputting each of the following GFF feature types: (gene, 5' UTR, CDS, exon, 3' UTR). These counts and lengths served as the input datasets for the analyses completed in Jupyter. After calculating the Reads Per Kilobase Million (RPKM) associated with each category using these counts and the feature lengths, we inferred the presence of intronic reads by taking the difference in these values between the annotated “gene” and “exon” categories. We then calculated the intron retention (IR) score for each gene using the following equation from IRFinder³:

$$IR = \text{Intron abundance} / (\text{Intron abundance} + \text{Properly spliced transcript abundance})^3$$

Statistical Analysis and Data Visualization: After performing kurtosis tests to determine whether the lengths and RPKM values associated with these different features follow a normal distribution, we performed the non-parametric Mann-Whitney U test to determine whether these characteristics are significantly different between transcripts with and without retained introns. We also compared the proportion of the total transcript length contributed by the 5' UTR, coding sequence (CDS), and 3' UTR sequences from the RPKM values. After analyzing the distribution of IR scores, we also generated a linear regression using the RPKM and length values for the different sequence components as our dependent variables. Statistical significance was determined using the alpha value 3.73e-06, calculated by dividing the p < 0.05 FWER by the total number of genes included in these datasets, as a more stringent cutoff to reduce the rate of false positives. Figures were generated using matplotlib.

RESULTS

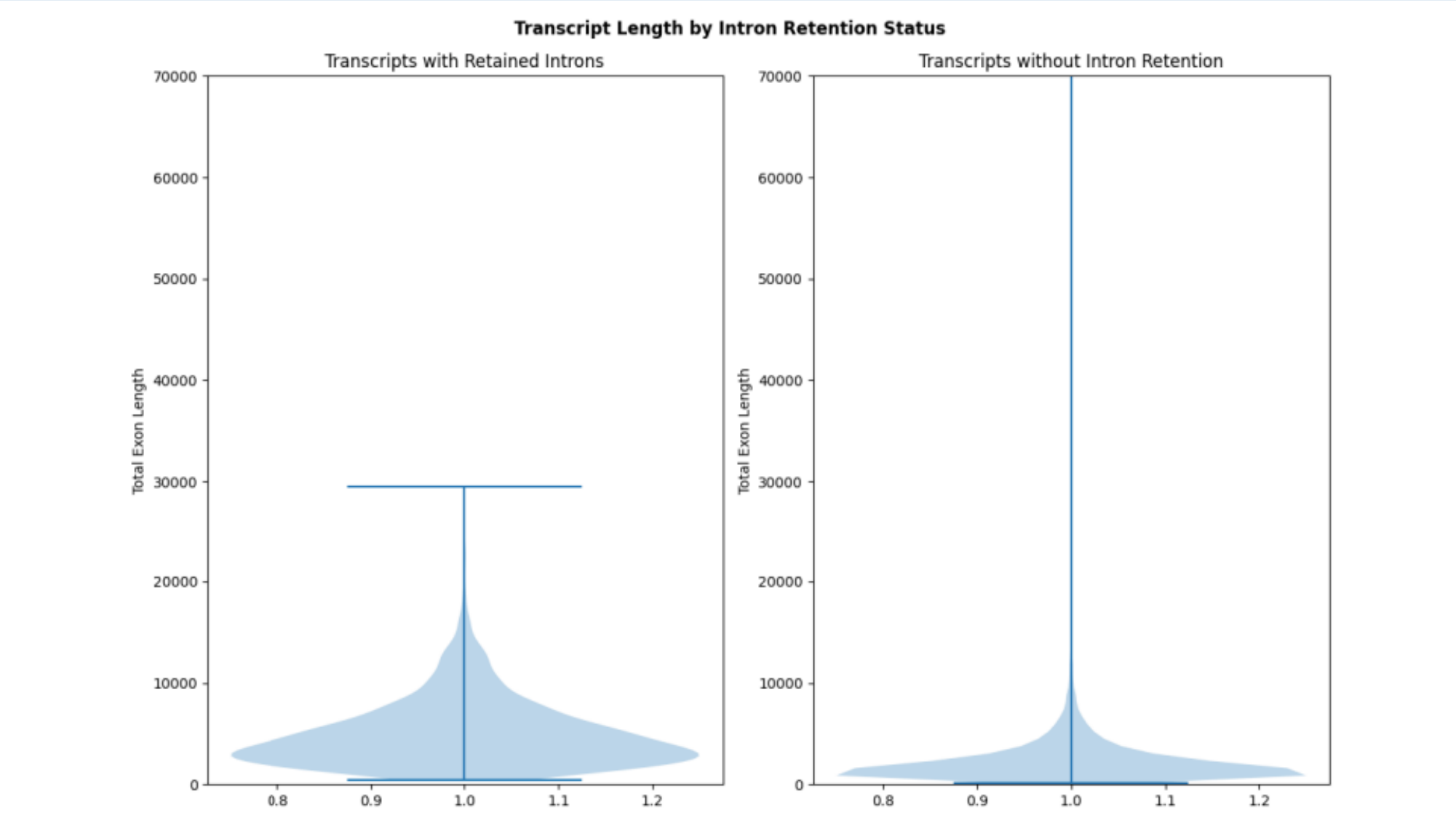


Figure 2: The range of overall transcript lengths is significantly wider for transcripts which do not include intron retention events. Figure 2 includes violin plots to visualize the distributions of overall sequence lengths for transcripts with (left) and without (right) intron retention events. There is a significant difference between the overall transcript lengths of these two groups (Mann-Whitney U test p-value = 0.0). This difference could be driven, at least in part, by a few individual outliers where the overall transcript length is incredibly long.

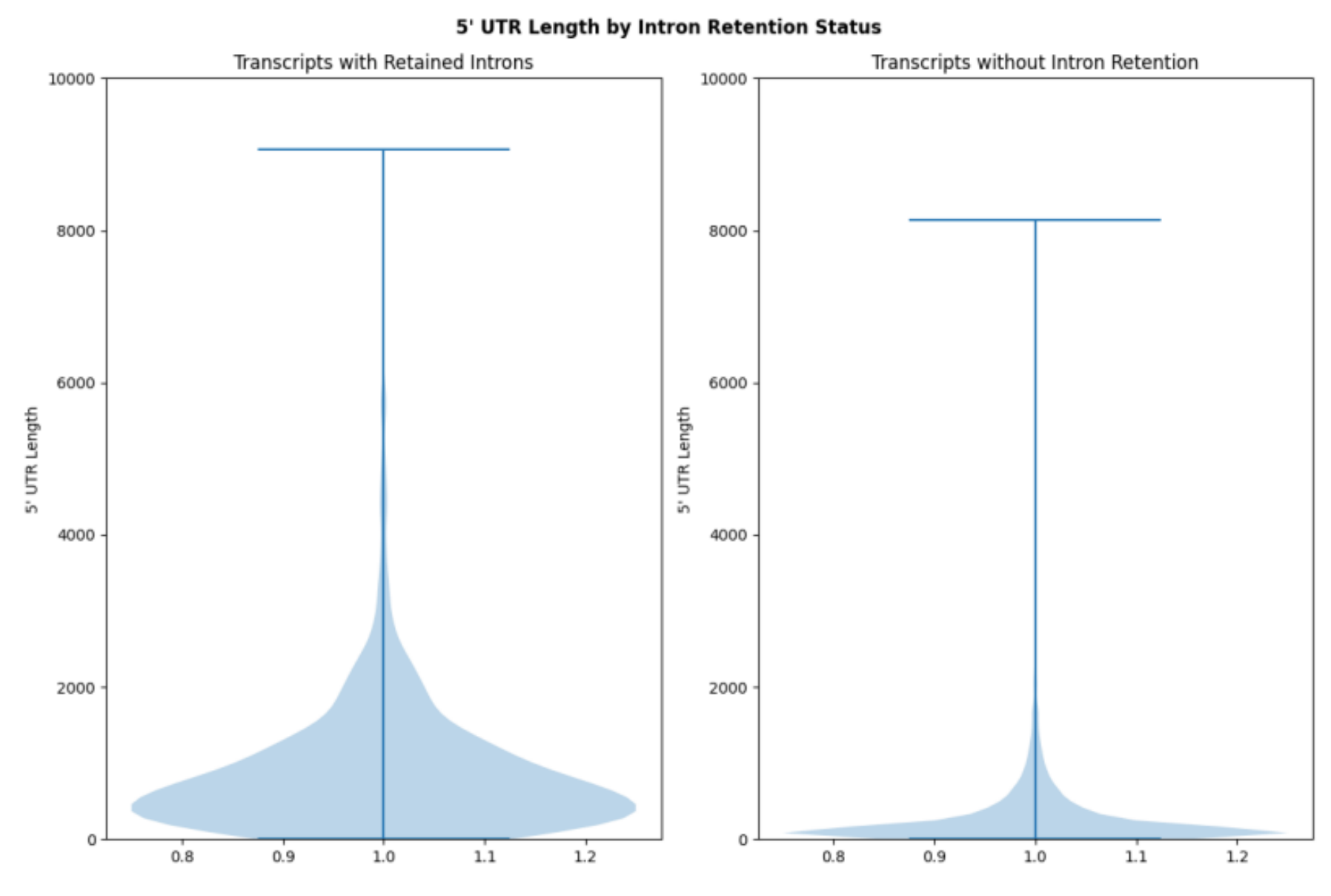


Figure 3: 5' UTR sequences appear to be longer in genes with retained introns than those without intron retention events. This figure displays violin plots comparing the distribution of 5' UTR sequence length for genes with transcripts that retain intronic sequences to those which do not. Results of the Mann-Whitney U test comparing these groups indicate that there is a significant difference between the mean 5' UTR lengths (p= 5.9e-255).

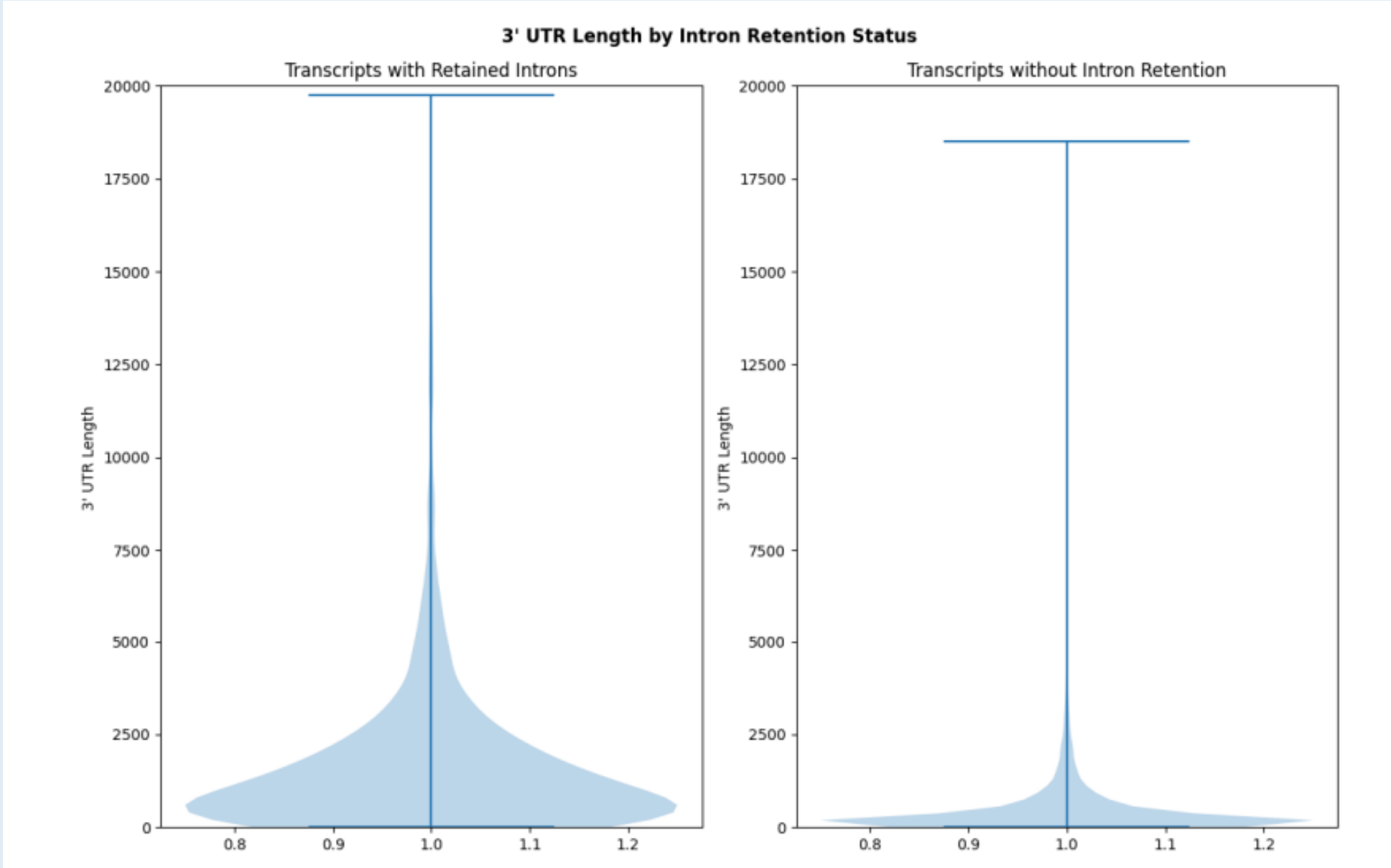


Figure 4: 3' UTR sequences appear to be longer in genes with retained introns than those without intron retention events. Figure 4 contains violin plots comparing the distribution of 3' UTR sequence length for genes with and without retained introns. Results of the Mann-Whitney U test comparing these groups indicate that there is a significant difference between the mean 3' UTR lengths (p= 1.42e-196).

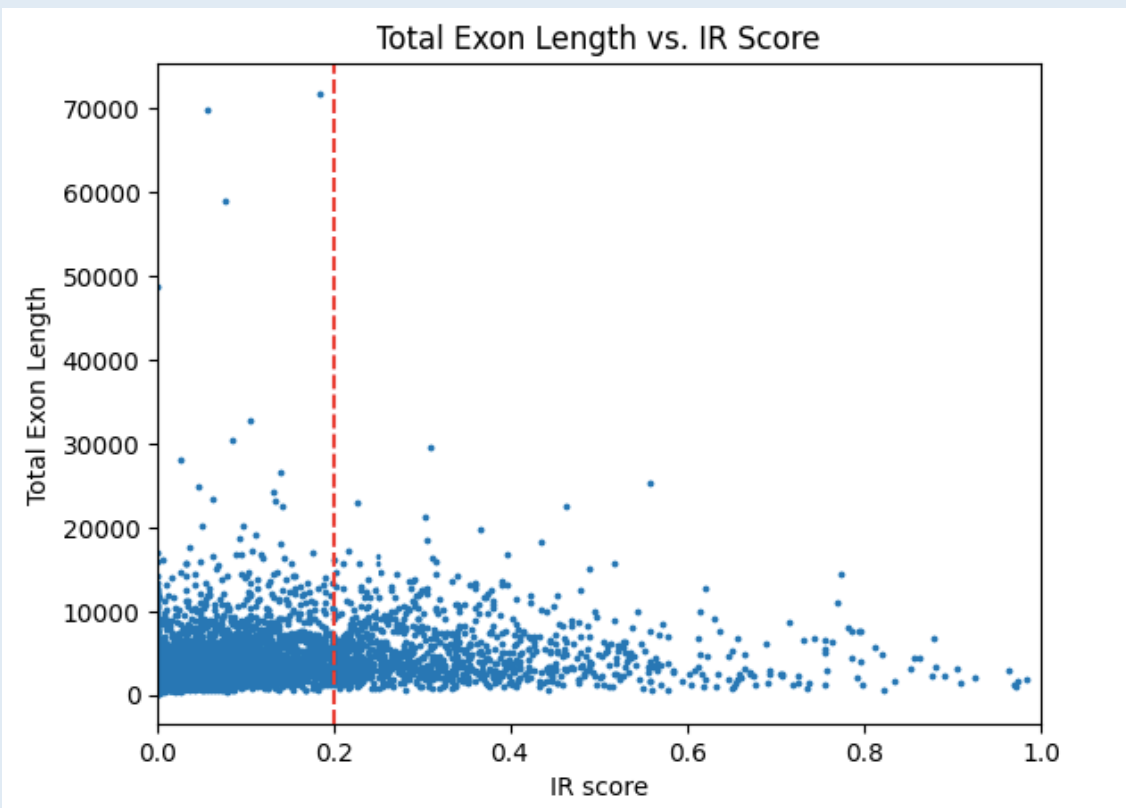
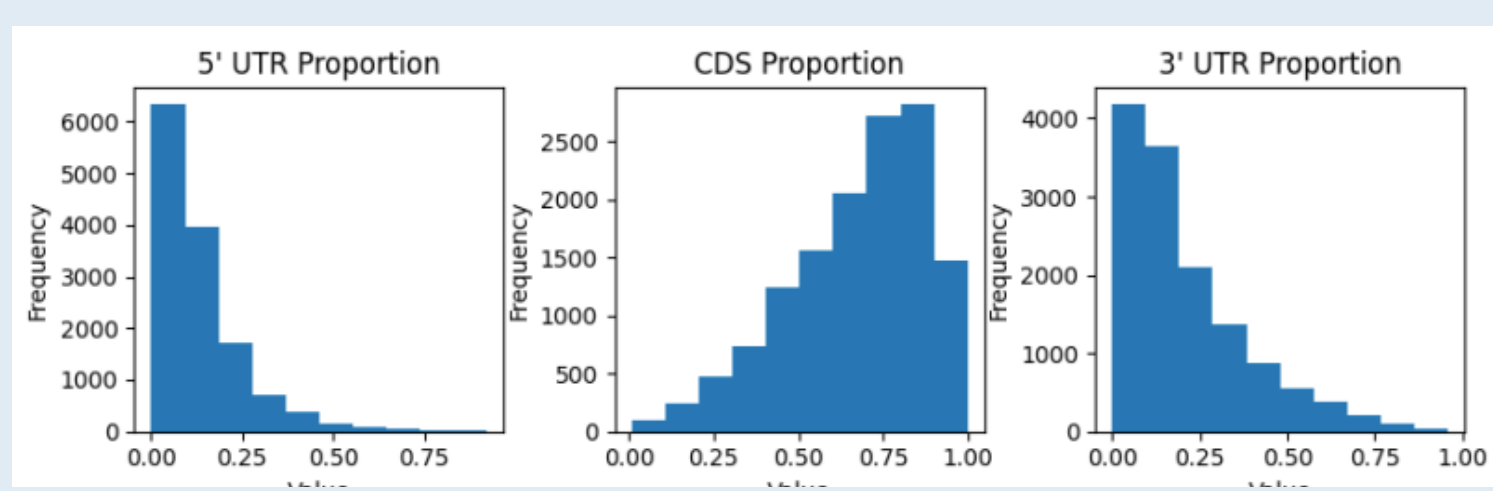


Figure 5: The Nab2 mutant fly transcriptome contains a wide range of intron retention scores. Figure 5 plots the total exon length against the intron retention (IR) score calculated for each gene. The red, dashed vertical line at x = 0.2 indicates the threshold for classifying whether transcripts of genes contain intron retention events. While many of the genes in each intron retention status category have similar transcript lengths, this plot also displays some of the longer sequences impacting the distributions of sequence lengths displayed in Figure 2.

Gene Name	Gene RPKM	5' UTR RPKM	CDS RPKM	3' UTR RPKM	Total Exon RPKM	Intron RPKM	Gene Length	5' UTR Length	CDS Length	3' UTR Length	Exon Length	IR Score	Intron Retention
1 Myo81F	0.040531	1.691231	3.109931	0.422808	3.004989	0.030667	1965857	195	6246	78	6519	0.756648	True
2 Parp	1.511686	49.688354	35.047095	14.390835	33.583046	0.697554	125704	75	2982	55	3112	0.461441	True
3 Alg-2	13.371431	138.600916	97.331269	16.392795	43.639648	1.386778	4548	74	534	682	1290	0.103712	False
4 Tim17b	9.759284	58.498457	144.624653	128.666229	94.578682	1.170996	10452	168	519	274	961	0.119988	False
5 CG41128	33.111508	264.179115	284.901882	88.981065	169.425062	2.614807	1991	95	216	53	364	0.078970	False
6 CG41099	8.005468	44.699754	44.545089	3.530389	34.909657	1.171620	21673	287	3376	766	4390	0.146352	False
7 CG42402	3.285652	9.805885	19.722595	8.982263	13.390997	0.887605	26388	1352	2341	1498	5061	0.270146	True
8 Gfat1	14.798896	23.139017	68.200744	98.246328	38.249286	1.425801	11441	2517	2191	191	4155	0.096345	False
9 CG40198	56.181930	159.005874	87.645267	55.432781	58.297268	0.000000	1626	28	552	987	1567	0.000000	False
10 CG12581	14.187499	58.157441	44.848128	18.921775	30.998203	0.401156	12120	761	1995	2705	5461	0.028275	False

Table 1: Information for the first 10 genes in the dataset includes the variables for analysis. Table 1 includes the first 10 rows of the table combining the RPKM and feature length values for the overall transcript, exon, 5' UTR, CDS, and 3' UTR sequence features to illustrate the data collected for each transcript in the Nab2 mutant female sample. Each row contains information for an individual gene's transcript. Overall, these data include 13404 genes, and 1097 of these genes have IR scores which are consistent with intron retention events (when defining the threshold as 0.2). Only the overall genes have significantly different RPKM values (p=9.44e-16) when we compare genes with and without retained introns and correct for multiple testing. We observed no significant differences in RPKM values for the 5' UTR, CDS, 3' UTR, or the exon sequences between the two groups of transcripts, based on our adjusted p-value cutoff. Differences in read levels, however, could occur if we compare individual genes' transcripts with and without the retained introns.



	5' UTR Proportion	3' UTR Proportion	CDS Proportion
count	13404.000000	13404.000000	13404.000000
mean	0.132441	0.210484	0.673444
std	0.117135	0.175200	0.202066
min	0.000287	0.000216	0.007806
25%	0.051403	0.079458	0.542271
50%	0.098470	0.154523	0.715357
75%	0.176609	0.295142	0.834320
max	0.922438	0.958062	0.999250

Figure 6: The proportions of each transcript defined as the 5' UTR, CDS, and 3' UTR sequences display different distributions. This panel includes both the histograms and descriptive statistics for the proportion of each transcript length annotated to the 5' UTR, CDS, and 3' UTR sequences.

CONCLUSIONS

- When considering the entire gene, rather than individual components of the sequence, genes with retained introns have significantly higher RPKM values than those which do not.
- Transcripts with retained introns appear to come from sequences with significantly longer components than those which do not contain retained introns.
- The proportion of the total transcript that includes the 5' UTR, and the proportion of the 3' UTR, are both skewed to the right, while the CDS proportion is skewed to the left. These proportions may also impact the frequency of intron retention events within these regions.

FUTURE DIRECTIONS

Potential future avenues of investigation include the following:

- Considering the location of the intron retention event and classify the transcripts based on the sequence components which contain them
- Comparing the intron retention events observed in this sequencing run to other genotypes (wild-type vs. mutant flies, males vs. females, etc.)
- Characterizing the relationship between the RNA modification N6-methyladenosine (m6A) and these intron retention events

ACKNOWLEDGEMENTS

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